Calin Cojocaru

Honours Student & Software Developer

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Education

Delft University of Technology **2024 – Present**
BSc Computer Science & Engineering | GPA: 8.17 *Delft, Netherlands*
Data Management, Machine Learning, Algorithm Design, Testing & QA, Big Data Processing

National College of Informatics "Tudor Vianu" **2020 – 2024**
High School Diploma | Baccalaureate: 10/10 in Math & CS *Bucharest, Romania*
National Olympiad Qualifier in Informatics (2024)

Professional Experience

Software Engineering Intern (Geolog International) **June 2026 – August 2026**
C# (.NET 10), EF Core, SQL Server, React, TypeScript, Bootstrap

- Developed a centralized web-based **Action Management System (AMS)** to replace fragmented communication, automating compliance lifecycles for 15+ complex QHSE forms across regional base locations.
- Engineered a modular, code-first backend with **EF Core** and **SQL Server**, implementing database-level global query filters to enforce country-level permission boundaries.
- Designed a schema-driven frontend utilizing **React**, **TypeScript**, and **Bootstrap**, minimizing client-side configuration overhead and reducing form deployment times for QHSE administrators.
- Implemented a hand-rolled workflow state machine in **C#** to automate multi-level approval lifecycles, system status transitions, and digital sign-offs with cryptographic hash-chaining.

Full-Stack Developer (Ador d.o.o.) **April 2026 – June 2026**
React, TypeScript, ASP.NET Core, PostgreSQL, Playwright

- Developed a full-stack AI-driven audit platform crawling client websites, running modular GDPR / SEO / accessibility checks, and generating reports via an **LLM**, eliminating manual reporting overhead and enabling reviewers to focus solely on compliance analysis.
- Implemented **static fetch path**, SPA detection heuristic, and the GDPR/cookie module.
- Designed the **REST API** and contributed to the ASP.NET Core backend with 70% branch coverage in **agile sprints**.

Technical Projects

Recommender System | Python, Jupyter, NumPy

- Engineered a dual User-User and Item-Item collaborative filtering pipeline handling high-dimensional sparse rating matrices.
- Implemented **Min-Hashing** to approximate Jaccard similarity, demonstrating key Locality-Sensitive Hashing (LSH) concepts for sub-quadratic similarity matching at scale.
- Applied **Non-negative Matrix Factorization (NMF)** for low-rank dimensionality reduction, achieving 82% accuracy using only raw NumPy linear algebra operations.

Grocery Product Classifier | Python, NumPy

- Built a fully modular ANN from scratch - linear layers, LReLU, Softmax, cross-entropy, full backprop; no AI frameworks - achieving ∼92% accuracy on a 7-class task.
- Diagnosed systematic class bias via confusion matrix analysis.

Brainfuck Interpreter | x86 Assembly (AT&T Syntax)

- Stack-based memory management with a **jump table** for constant-time instruction decoding, replacing branching logic and reducing execution time by over 80%. Developed on Linux (Ubuntu)

Technical Stack & Competencies

Programming Languages: C++, Python, C#, Assembly, Java, Scala, SQL, JavaScript, Unix
Tools: Visual Studio, PyCharm, IntelliJ, CLion, Linux (CLI), Git
Competencies: LLM Integration, Machine Learning, Neural Networks, Algorithm Design, Computer Graphics, OOP Design Patterns, Unit Testing & Mocking, Agile/Sprint Workflows, Bash Scripting
Spoken Languages: Romanian (Native), English (Fluent), German (B1)
Certifications: Cisco CCNA-1 (Introduction to Networks)